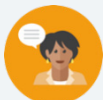


3.2 Rational Numbers and the Number Line – Part 2

Lesson Introduction

 Benchmark	MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.		 Learning Targets	- I can determine how far a number and its opposite are from zero. - I can represent a number and its opposite on a number line. - I can compare a quantity and its opposite on a horizontal or vertical line and explain the meaning of zero within context.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is the meaning of a number’s opposite? - How can you use tick marks between each value to determine the locations of rational numbers on a number line? - How is zero used to determine where to plot a number and its opposite on a number line?			
 Lesson Narrative	In this lesson, students will work on developing their understanding of a number and its opposite. During the lesson, students will use a number line to represent a number and its opposite, showing that they are the same distance from zero on a number line. Students also use the number line to compare numbers and their opposites.			
 Component Summary	3.2.1 Warm-Up (~5 min.)	Notice and wonder about words that are opposites (<i>SE p. 82</i>)	3.2.4 Try It (5 min.)	Plot opposites on horizontal and vertical number lines (<i>SE p. 84</i>)
	3.2.2 Activity (10 min.)	Explore the relationship between opposites on number lines (<i>SE p. 83</i>)	3.2.5 Guided Instruction (5 min.)	Compare numbers and their opposites using greater and less than symbols (<i>SE p. 85</i>)
	3.2.3 Collaboration (10 min.)	Plot numbers and their opposites on horizontal number lines (<i>SE p. 83</i>)	3.2.6 Your Turn! (5-10 min.)	Plot values on a number line and compare their values using greater and less than symbols (<i>SE p. 86</i>)
	3.2.7 Wrap-Up (~5 min.)	Describe the relationship between opposites and zero on a number line (<i>SE p. 86</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Rational numbers - Opposites		- Paper to create foldable - Number line from Lesson 1	
			Additional Preparation	
			(Optional) For 3.2.2 Activity, prepare labeled number lines to save time in the lesson.	

 Teacher Tips	Common Misconceptions - Students may struggle with plotting negative fractions and decimals when using irregular intervals, particularly in 3.2.4 Try It. - Students may get rational numbers and integers confused in 3.2.3 Collaboration. - Students may think inequality symbols can only be read in one direction. For example, they may not understand $a < b$ can be interpreted as “a is less than b,” and also as “b is greater than a.”	Things to Consider - Use the number line created to show that when folded at zero, opposite numbers will overlap, showing that they are the same distance from zero. - Use different colors to help students find the distance from zero. - In 3.2.2 Activity, describe what a vertical crease is. - Consider using graph or lined paper to create the foldable number lines to ensure equally spaced tick marks.
	Literacy and Language Routines Notice and Wonder In 3.2.1 Warm-Up, students notice and wonder about familiar words that are opposites. This activity develops students’ conceptual understanding of opposite numbers, as they are able to connect the new concept to vocabulary terms they may already be familiar with.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection In 3.2.3 Collaboration, students analyze and critique the thinking of others. They articulate their thinking as they discuss their analysis of the samples of student work.
	 Differentiate Instruction 3.2.2 Activity provides kinesthetic and visual entry points as students continue developing their understanding of negative values. Get creative and use real-world examples like a football field or the human wingspan to engage students.	
Lesson Notes:		

3.2.1 Warm-Up: Word Pairs

Component Overview: Students develop a conceptual understanding of opposites by connecting the concept of opposite to real-world vocabulary they may be familiar with. Students are asked what they notice and wonder about the pairs of words.



3.2.1 Warm-Up: Word Pairs

Up	Down
Left	Right
Clockwise	Counterclockwise
Increase	Decrease
Ebb	Flow
West	East
North	South
Gain	Lose



This activity can be facilitated as a Notice and Wonder routine. Students do not need to identify the pairs of words as opposites, but if they do, use it as a bridge to the math application of opposites.

1. What do you notice about the pairs of words above?

Answers will vary. They are opposites. They involve direction.

2. How do you think this may be used in the study of mathematics?

Answers will vary. Opposites could be represented with positive and negative values.

3.2.2 Activity: Folded Number Line

Component Overview: Students explore opposites on a number line. If using a sheet of paper as a foldable, ensure students crease the paper exactly on the zero of their number lines. After folding, have students discuss the relationship between numbers and their opposites relative to zero on the number line.



3.2.2 Activity: Folded Number Line

On a separate sheet of paper, create a number line with integers from -10 to 10 to answer the following questions. Fold the number line so that the vertical crease goes through 0 and the two sides of the number line match up perfectly.

- When the number line is folded, which number is the same distance away from 0 as the number 4 ?
 -4
- When the number line is folded, which number is the same distance away from 0 as the number -7 ?
 7
- When the number line is folded, what do you notice about the numbers that are the same distance away from 0 ?
Answers will vary. They are the same number, but one is positive and one is negative.
- Find another pair of numbers that are the same distance away from 0 .
Answers will vary. 1 and -1 . 9 and -9 .
- Two numbers that are the same distance from zero on the number line are called opposites.



If you have markers that bleed, consider having students draw and label only half of the number line or only the positive numbers. Then fold in half. If the marker bleeds well, the students will unfold to reveal a mirror image of the number line. Have students then correctly label those values as negative, identify opposites, and describe their relationship in terms of distance from zero.



What do you notice about the location of opposites on a number line? Is this always true? Why or why not?

3.2.3 Collaboration: Opposites

Component Overview: Students choose their own two values, one positive and one negative, for W and F . Students will therefore have a variety of answers. This activity is intentionally scaffolded, with each question acting as a natural question stem and next step for understanding opposites. Students should complete the entire activity with their partner before debriefing any answers.



3.2.3 Collaboration: Opposites

With your partner, choose values for W and F . The value for W can be any rational number between -8 and 0 that is not an integer. The value for F can be any rational number between 0 and 8 that is not an integer.

Answers will vary.

The value for W is -5.2 . The value for F is 3.5 .

- Plot and label both numbers on the number line.

Answers will vary based on choice of W and F .



- Plot the opposite of W . Label the point X .

Answers will vary based on choice of W . Answer shown on number line above.

- What is the value of point X ?

Answers will vary based on choice of W . 5.2

- How far is point X from 0 ?

Answers will vary based on choice of W . 5.2 units

- Plot the opposite of F . Label the point G .

Answers will vary based on choice of F . Answer shown on number line above.

- What is the value of point G ?

Answers will vary based on choice of F . -3.5

- How far is point G from 0 ?

Answers will vary based on choice of F . 3.5 units

- Noah said, "The expression $-a$ can be read 'negative a ' or 'the opposite of a '. So, if a is a rational number, $-a$ will always be a negative number."

Use points F , G , W , and X to explain to Noah that the opposite of a value could be positive.

Answers will vary. The opposite of a positive number will always be negative. The opposite of a negative number will always be positive. The expression " $-G$ " can be read "negative G " or "the opposite of -3.5 " which is positive 3.5.



This activity is intentionally scaffolded to guide students' thought processes for discovering a number and its opposite's relationship, relative to their distance from zero on a number line. Review the 3.2.7 Wrap-Up question and determine framing questions to use here to best prepare students for the Wrap-Up.



Is Noah correct? Why or why not? Can you have an opposite of an opposite? What would that look like represented mathematically? Does "opposite" always mean "negative"?

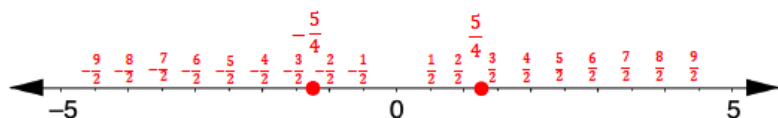
3.2.4 Try It: Opposites

Component Overview: Students apply skills of plotting opposites on both horizontal and vertical number lines. Students compare values, identifying which is larger. Students should recognize that positive values are always greater in value than negative values.



3.2.4 Try It: Opposites

- Use the number line below to answer the following problems.



- Label the tick marks on the number line. Then plot the value $\frac{5}{4}$ and its opposite on the number line.
- Which value is larger, $\frac{5}{4}$ or its opposite?

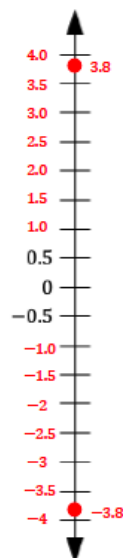
$$\frac{5}{4}$$

- Use the vertical number line to answer the following problems.

- Label the tick marks on the number line. Plot and label the value -3.8 and its opposite on the number line.

- Which value is larger, -3.8 or its opposite?

3.8 is larger.



Consider using a Think Aloud for a small group of students that need additional support.

- How do you determine how to label a number line? What scale do you use?
- What is the range of the part you are labeling ($0 - 5$)?
- How many tick marks are there (10)?
- So how could you divide five into 10 equal parts?
- What would the value of each part be if 10 equal parts equaled five? ($\frac{1}{2}$ or 0.5)



Students do not have to label every tick mark if they do not need to do so to determine the location of each value on the number lines. For example, they could label only the integer values, if they wish, since this would not affect the placement of the points on the number lines.

3.2.5 Guided Instruction: Comparing a Quantity and Its Opposite

Component Overview: Students review how to use less than and greater than symbols to compare two values. Students finish the activity by reasoning about the values of numbers and their opposites, relative to their position on a number line and whether they are positive or negative.

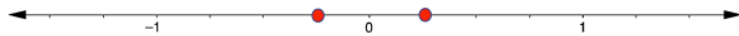


3.2.5 Guided Instruction: Comparing a Quantity and its Opposite

Plot each pair of values on the number line. Choose an answer to correctly fill in the blank.

1.

- a. Plot -0.25 and 0.25 .

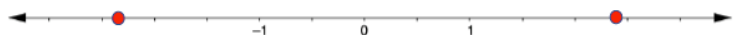


- b. The value -0.25 is ☐ greater ☒ less than 0.25 .

- c. -0.25 ☐ $>$ ☒ $<$ 0.25

2.

- a. Plot $2\frac{2}{5}$ and $-2\frac{2}{5}$.



- b. The value $2\frac{2}{5}$ is ☐ greater ☒ less than $-2\frac{2}{5}$.

- c. $2\frac{2}{5}$ ☐ $>$ ☒ $<$ $-2\frac{2}{5}$

3. In each comparison, the number on a horizontal number line that is farther to the ☐ right ☒ left is the greater value.

4. When comparing opposites, the ☐ positive ☒ negative value will always be the greater value.



Have students practice interpreting these mathematical statements, particularly reading the same statement in multiple ways. For example, “ -0.25 is less than 0.25 , and 0.25 is greater than -0.25 .” This will support students’ conceptual understanding of inequality statements.

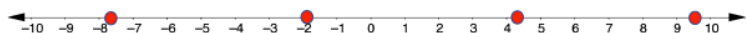
3.2.6 Your Turn!

Component Overview: Students continue practicing plotting, ordering, and comparing values of rational numbers. Circulate while students work to support struggling students, monitor student progress, and select students that you want to have share during the debrief.



3.2.6 Your Turn!

- Plot and label each value and its opposite on the number line: 9.5, -2, 4.2, $-7\frac{3}{4}$.



- Pairs of opposite numbers are ☐ the same ☐ different distance from zero on a number line.

- Complete each comparison statement for the pairs of numbers you plotted in problem #1. For each statement, use < or > in the first blank and the opposite of the given number in the second blank.

a. 9.5 > -9.5

b. -2 < 2

c. 4.2 > -4.2

d. $-7\frac{3}{4}$ < $7\frac{3}{4}$



For students who might finish early, consider challenging them by beginning to compare values other than opposites. In the following lessons, students begin comparing the values of rational numbers.



What is your strategy for plotting fractions and decimals? How do you determine which integers to plot it between and how close or far to plot from an integer?



Consider providing guiding questions for students who struggle with plotting decimals and fractions. For example, if plotting 4.2, students can ask themselves questions like:

- What is the closest whole number?
- Is it greater or less than that number?
- Does that mean it will be to the right or left of it on the number line? How far away?

3.2.7 Wrap-Up: Cool Down

Component Overview: This Wrap-Up evaluates students' understanding of opposites and their locations on the number line.



3.2.7 Wrap-Up: Cool Down

1. What is the relationship between a number and its opposite? Use the words **number line** and **zero** in your explanation.

Answers will vary. On a number line, a value and its opposite are the same distance from zero, but on opposite sides of zero. For example, the opposite of 4 is -4 . Both 4 and its opposite, -4 , are 4 units away from zero on a number line.



For students who might struggle with communicating their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.